Spot-Spray Weed Mapping — Technical Documentation

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Overview

The project detects weed patches in high-resolution RGB imagery from drones, turns them into sprayer-ready treatment zones, and estimates how much herbicide a spot-spray pass would save compared with spraying the whole field.

Scope

- In: RGB orthomosaics (or georeferenced frames) of row crops or fallow fields; field boundaries; optional crop row direction and sprayer specs.
- Out: weed detection maps, spray zone polygons and prescription files, savings estimates with uncertainty, and an accuracy report.
- Not in scope: species-level herbicide choice, real-time on-boom detection, and multispectral or hyperspectral inputs (v1 is RGB only).

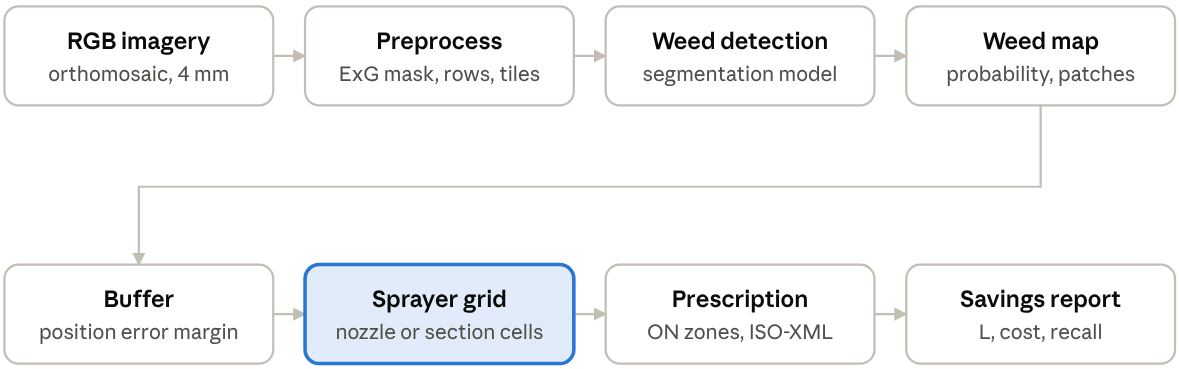
Key outputs

Output	Format
Weed probability raster	Cloud-Optimized GeoTIFF
Weed instances (boxes/masks)	GeoPackage
Spray zones (buffered, sprayer-aligned)	GeoPackage, Shapefile, ISO-XML / ISOBUS prescription
Savings report	JSON + HTML/PDF
Accuracy report	JSON + HTML

Pipeline

Detection produces a weed map; savings come only after that map is converted to what a real sprayer can switch, which is where most of the gap between "weed area" and "herbicide saved" lies.

Savings are computed on the sprayer grid, not on raw detections



processing pipeline · imagery to savings

Stack: Python 3.11, rasterio/GDAL, OpenCV, PyTorch with segmentation-models-pytorch (or Ultralytics for detection), geopandas/shapely for zones, and a Typer CLI `weedmap`. Detection runs on GPU; zones and savings run on CPU in seconds per field.

Imagery requirements and preprocessing

Detection quality depends mainly on ground sample distance and timing: small weeds must span enough pixels, and imagery must be captured when weeds are present but before canopy closure hides them.

Parameter	Requirement	Why
Ground sample distance	⩽ 5 mm/px for early seedlings; ⩽ 1 cm/px for established patches	A 2 cm weed at 5 mm is ~4 × 4 px, near the detection floor
Overlap	75% front / 70% side for orthomosaics	Clean stitching over uniform crop rows
Lighting	Overcast or high sun; avoid low-angle shadows	Shadows read as dark vegetation and confuse colour cues

Parameter	Requirement	Why
Timing	Pre-emergence burndown or early post-emergence (crop ⩽ ~V4/4-leaf)	Weeds visible; crop not yet covering inter-row
Positioning	RTK/PPK or ⩾ 5 GCPs; ⩽ 5 cm absolute error	Spray zones must land on the weeds
Motion blur	Shutter ⩾ 1/1000 s	Fine leaf edges matter for small targets

Preprocessing

1. **Orthomosaic** — built externally (e.g. OpenDroneMap, Pix4D, Metashape); the pipeline also accepts raw georeferenced frames for higher effective resolution.
2. **Colour normalization** — white-balance per flight against a grey card or with gray-world correction; per-tile histogram matching to the training distribution.
3. **Vegetation pre-mask** — Excess Green (ExG = 2G − R − B, on chromatic coordinates) with Otsu thresholding separates vegetation from soil and residue; non-vegetation is skipped, cutting inference time.
4. **Crop row detection** — Hough transform on the vegetation mask estimates row direction and spacing; used to flag inter-row vegetation and to align spray zones with sprayer travel.
5. **Tiling** — 1 024 × 1 024 px tiles with 128 px overlap; tile footprints stored for stitching.

Weed detection

The default model is semantic segmentation (crop / weed / soil per pixel), because spray decisions need weed *area* and location, not just counts; an object detector is offered for sparse, isolated weeds.

Approaches

Approach	Model	Best for
Rule-based baseline	ExG mask + "vegetation not on a crop row"	Pre-emergence burndown (any green = weed); quick start with no labels

Approach	Model	Best for
Semantic segmentation (default)	U-Net or DeepLabv3+ with ResNet/EfficientNet encoder; SegFormer-B2 as alternative	Patches and mixed in-row weeds
Object detection	YOLO-family detector (boxes or instance masks)	Sparse individual weeds, counts per m²

Labels — classes: `soil`, `crop`, `weed` (optional sub-classes: grass vs broadleaf, since herbicide choice often splits that way). Annotate polygons in CVAT or Label Studio on 512 × 512 crops; target ≥ 300 labelled crops across ≥ 5 fields, growth stages and lighting conditions. Public datasets (e.g. crop/weed segmentation sets for sugar beet, maize, soybean) can be used for pre-training; check licences.

Training

- Loss: Dice + cross-entropy (or focal) to handle weed pixels being a small share of the image.
- Augmentation: flips, 90° rotations, scale ±25%, brightness/contrast/hue jitter, blur, simulated shadows.
- Split by **field**, not by tile: tiles from one field never appear in both train and test, otherwise neighbouring tiles leak.
- Metrics: weed IoU, weed precision/recall at pixel and patch level; model selection on validation weed F1.
- Inference: sliding window with overlap and logit averaging; test-time augmentation optional.

Post-processing — threshold the weed probability (default 0.5, tunable for the cost trade-off below), remove components below `detect.min_area_cm2` (default 4 cm²), and suppress weed pixels inside the crop row band when the confidence margin over `crop` is small.

From detections to spray zones

A sprayer cannot hit a 3 cm weed exactly, so detections are converted into zones at the resolution the target sprayer can actually switch: per nozzle, per boom section, or per grid cell.

Steps

1. **Safety buffer** — dilate each weed region by `zones.buffer_m` (default 0.25 m) to cover positioning error, latency and unseen nearby weeds.
2. **Sprayer grid** — overlay a grid aligned with the travel direction (from crop rows or a given heading): cell width = nozzle spacing × nozzles per control unit; cell length = travel speed × valve response time + minimum on-time distance.
3. **Cell decision** — a cell is ON when its weed area ≥ `zones.min_weed_cm2` (default 1 cm², i.e. any weed) or, for patch-level management, when weed cover ≥ `zones.cover_threshold` %.
4. **Smoothing** — fill single OFF cells between ON cells along travel, and drop isolated ON runs shorter than the minimum on-time, matching valve behaviour.
5. **Export** — ON cells dissolved into polygons with a `rate` attribute (full rate or 0) as a prescription map.

Sprayer presets

Preset	Control unit width	Typical use
`nozzle`	0.25–0.5 m	PWM individual-nozzle control
`section`	3–6 m	Section control on a conventional boom
`drone`	2–5 m swath	Spray drone with on/off by swath
`grid`	user-defined (e.g. 1 m, 5 m)	Research and comparison

Verify actual widths and response times against the user's equipment; they drive savings more than the detector does.

Herbicide savings estimation

Savings are the share of field area the sprayer would leave OFF, converted to product volume and cost, and reported as a range that accounts for detection errors rather than a single number.

$$S = 1 - \frac{A_{\mathrm{sprayed}}}{A_{\mathrm{field}}}, \qquad V_{\mathrm{saved}} = S \cdot A_{\mathrm{field}} \cdot R, \qquad C_{\mathrm{saved}} = V_{\mathrm{saved}} \cdot P$$

where $A_{sprayed}$ = area of ON cells after buffering and smoothing, R = label application rate (L/ha), P = product price per litre. Spot spraying sometimes uses a higher rate per treated area; `savings.spot_rate_multiplier` (default 1.0) captures this.

Report contents

- Field area, treated area and savings % for each sprayer preset, side by side — showing how much is lost to coarser control units.
- Volume (L) and cost saved, per field and in total; carrier water saved.
- Sensitivity curve: savings vs detection threshold, with weed recall on the same axis, so the user sees the cost of missing weeds.
- Weed pressure: weed cover % and patch count per hectare.

Uncertainty — missed weeds (false negatives) make savings look better than real-world efficacy; false positives make them look worse. Using validation precision and recall at the chosen threshold, the treated area is bootstrapped (resampling validation fields) to give a 90% interval on S. Report weed recall next to every savings figure.

Agronomic caveat — high savings are only valid if efficacy holds. A missed-weed rate is listed alongside, and the report recommends a full-coverage pass when weed cover exceeds `savings.blanket_above_cover` (default 30%), where spot spraying saves little.

Configuration, CLI and outputs

One YAML file per field campaign holds imagery paths, model choice, sprayer preset and product prices, so savings can be recomputed for a different sprayer without re-running detection.

`config/campaign.yaml`

```
campaign: 2026_soy_postemerge
fields: data/fields.gpkg
imagery: data/ortho/{field_id}.tif
gsd_m: 0.004
detect:
  model: models/unet_effb3_v2.pt     # or: rule_based | yolo_v2.pt
  tile_px: 1024
  overlap_px: 128
  threshold: 0.5
  min_area_cm2: 4
zones:
  heading: auto                       # auto = from crop rows, or degrees
```

```
  buffer_m: 0.25
  presets: [nozzle, section]
  nozzle: {width_m: 0.5, min_on_m: 0.5}
  section: {width_m: 3.0, min_on_m: 2.0}
  min_weed_cm2: 1
savings:
  products:
    - {name: glyphosate_540, rate_l_ha: 1.8, price_per_l: 6.50}
  carrier_l_ha: 100
  spot_rate_multiplier: 1.0
  blanket_above_cover: 30
```

Prices and rates above are placeholders; enter the label rates and local prices that apply.

CLI

```
weedmap preprocess --config config/campaign.yaml
weedmap detect     --config config/campaign.yaml --field F07
weedmap zones      --config config/campaign.yaml --preset nozzle
weedmap savings    --config config/campaign.yaml
weedmap evaluate   --config config/campaign.yaml --labels
data/val_labels.gpkg
weedmap train      --data data/labels/ --arch unet --encoder efficientnet-b3
```

Output layout

```
out/<campaign>/<field_id>/
  weed_prob.tif  weeds.gpkg
  zones_<preset>.gpkg  rx_<preset>.zip (ISO-XML / shapefile)
  savings.json  report.html
```

Validation, limitations and roadmap

The headline number is savings at a stated weed recall, validated on fields the model never saw, and ideally confirmed with a field strip trial.

Validation

- **Detection** — held-out fields: pixel IoU and F1 for `weed`; patch-level recall (share of hand-labelled weed patches touched by a detection) at 0, 10 and 25 cm tolerances.
- **Zones** — recall of labelled weeds inside ON zones per sprayer preset; this is the number that matters for efficacy.

- **Ground truth** — quadrat counts (e.g. 0.25 m² frames at 30+ georeferenced points per field) to check that imagery-based weed cover matches in-field counts, including weeds hidden under crop.
- **Field trial (recommended)** — paired strips sprayed blanket vs from the prescription; compare product used (sprayer logs) and weed control at 14–21 days after treatment.

Known limitations

- RGB struggles with weeds that look like the crop (grass weeds in cereals, volunteer crop) and with weeds under canopy.
- Very small seedlings (< 1 cm) are below the detection floor at typical drone GSD.
- Imagery-to-spray delay lets new weeds emerge; savings assume the map is still current at spray time.
- Savings depend heavily on sprayer control width and response time, which are inputs, not measured.
- Residual (pre-emergence) herbicides need full coverage and are not candidates for spot spraying.

Roadmap

- [ ] Grass vs broadleaf classification for split-product prescriptions
- [ ] Multispectral input option (NIR/red-edge) for better crop–weed separation
- [ ] Active learning loop to pick the most informative tiles for labelling
- [ ] Direct export to common terminal formats with per-nozzle maps